Complete at least 2 exercises.

Throughout, $\Bbbk$ is a field and all modules are graded right modules. A graded algebra $A = \bigoplus_{i\geq 0} A_i$ is locally finite-dimensional with $S = A_0$ semisimple; in all explicit examples $S = \bigoplus_i \Bbbk e_i$ with coordinate idempotents e_i.

Exercise 1.

(1) Let $B = T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)/(x^2)$ with $\deg x = \deg y = 1$.

i. Let $a_d = \dim_{\Bbbk} B_d$. Prove that $a_d = a_{d-1} + a_{d-2}$ for $d \geq 2$ and $h_B(t) = \dfrac{1+t}{1-t-t^2}$.

Solution: A basis of B_d is the set of words in x, y of length d containing no xx (the ideal (x^2) is spanned by words containing xx). For $d \geq 2$: words starting with y are $y\cdot w$ with w admissible of length $d-1$ (a_{d-1} choices); words starting with x must continue with y, and are $xy\cdot w$ with w admissible of length $d-2$ (a_{d-2}). So $(1-t-t^2)h_B = a_0+(a_1-a_0)t = 1+t$.

ii. Compute $B^!$.

Solution: Let $X = x^*$ and $Y = y^*$. For $F = \alpha XX + \beta XY + \gamma YX + \delta YY$ one has $F(x \otimes x) = \alpha$, so $R^\perp = \operatorname{span}\{XY, YX, YY\}$ and $B^! = \Bbbk\langle X, Y\rangle/(XY, YX, Y^2)$.

iii. Compute $h_{B^!}(t)$.

Solution: Any word of length ≥ 2 containing Y contains one of XY, YX, YY; so $\{X^n \mid n \geq 0\} \cup \{Y\}$ spans, and it is a basis because the ideal is spanned by words containing XY, YX or YY, none of which occurs in X^n or Y. Thus $h_{B^!}(t) = \frac{1}{1-t} + t$.

(2) For $q \in \Bbbk$ let $C_q = \Bbbk\langle x, y\rangle/(x^2,\ y^2,\ xy + qyx)$ with $\deg x = \deg y = 1$.

i. Prove that $C_q \cong C_{q'}$ as graded algebras if and only if $q = q'$ or $qq' = 1$.
Hint. Consider the basis $\{1, x, y, yx\}$ of C_q. It may be helpful to consider $u^2 = \alpha\beta(1-q)\,yx$ for $u = \alpha x + \beta y$ with $\alpha, \beta \in \Bbbk$.

Solution: ($\Leftarrow$) If $qq' = 1$ then $q \neq 0$ and $x \mapsto y$, $y \mapsto x$ sends x^2, y^2 to y^2, x^2 and $xy + qyx \mapsto yx + qxy = q(xy + q'yx)$, so it induces a graded isomorphism $C_q \to C_{q'}$ (inverse of the same form). ($\Rightarrow$) Let $\varphi : C_q \to C_{q'}$ be a graded isomorphism; φ is determined by the bijection φ_1 on degree 1 and preserves $u^2 = 0$. If $q = 1$, every u squares to zero in C_1, hence in $C_{q'}$, so $q' = 1$; symmetrically $q' = 1 \Rightarrow q = 1$. If $q, q' \neq 1$, then φ_1 permutes the two lines $\Bbbk x, \Bbbk y$ (the only u with $u^2 = 0$): either $\varphi(x) = \lambda x$, $\varphi(y) = \mu y$, and $0 = \varphi(xy + qyx) = \lambda\mu(q - q')yx$ gives $q = q'$; or $\varphi(x) = \lambda y$, $\varphi(y) = \mu x$, and $0 = \lambda\mu(yx + qxy) = \lambda\mu(1 - qq')yx$ gives $qq' = 1$ (here $\lambda\mu \neq 0$ and $yx \neq 0$).

ii. Deduce that $C_1 \ncong C_{-1}$ if $\operatorname{char}\Bbbk \neq 2$.

Solution: Now if $\operatorname{char}\Bbbk \neq 2$, then $q = 1$, $q' = -1$ satisfy neither $q = q'$ nor $qq' = 1$, so $C_1 \ncong C_{-1}$.

iii. Compute a quadratic presentation of $C_q^!$.

Solution: Let $X = x^*$, $Y = y^*$. For $S = \Bbbk$, $(\lambda \otimes \mu)(v \otimes w) = \mu(v)\lambda(w)$. For $F = \alpha X \otimes X + \beta X \otimes Y + \gamma Y \otimes X + \delta Y \otimes Y$,

$$F(x \otimes x) = \alpha, \qquad F(y \otimes y) = \delta, \qquad F(x \otimes y + q\, y \otimes x) = \gamma + q\beta.$$

So $R^\perp = \Bbbk(X \otimes Y - qY \otimes X)$ and $C_q^! = \Bbbk\langle X, Y\rangle/(XY - qYX)$.

iv. Compute $h_{C_q^!}(t)$.

Solution: Using $XY = qYX$, every word is a multiple of $Y^a X^b$, so these span; they are independent: for $q = 0$ the relation is monomial and $\{Y^a X^b\}$ is the set of words avoiding XY; for $q \neq 0$ let X act on $\Bbbk[s,t]$ by multiplication by s and Y by $(Yf)(s,t) = t\, f(q^{-1}s, t)$; then $XY = qYX$ and $Y^a X^b \cdot 1 = q^{-ab} s^b t^a$, which are independent. Hence $h_{C_q^!} = \sum_d (d+1)t^d = (1-t)^{-2}$.

v. Show that $C_0^!$ and $C_1^!$ has the same Hilbert series while being non-isomorphic algebras.

Solution: $C_1^! = \Bbbk[X, Y]$ is commutative while in $C_0^! = \Bbbk\langle X, Y\rangle/(XY)$ one has $XY = 0 \neq YX$: same Hilbert series, non-isomorphic algebras.

(3) Let $P_\bullet \to M$ be a projective resolution of an ordinary (ungraded) module with differentials d_h and let $K_1 = \operatorname{Ker} d_0$.

i. Prove that

$$\operatorname{Ext}_A^1(M, N) \cong \operatorname{coker}\big(\operatorname{Hom}_A(P_0, N) \to \operatorname{Hom}_A(K_1, N)\big),$$

where the map is restriction.

Solution: Let δ^h be the differentials of $\operatorname{Hom}_A(P_\bullet, N)$. For $h = 1$: $\operatorname{Ker}\delta^1$ consists of maps $P_1 \to N$ vanishing on $\operatorname{Im} d_2 = \operatorname{Ker} d_1$, that is, maps factoring uniquely through $d_1 : P_1 \twoheadrightarrow K_1$. So $\operatorname{Ker}\delta^1 \cong \operatorname{Hom}_A(K_1, N)$, and under this identification $\operatorname{Im}\delta^0 = \{\varphi d_1 : \varphi \in \operatorname{Hom}_A(P_0, N)\}$ is the image of the restriction map $\operatorname{Hom}_A(P_0, N) \to \operatorname{Hom}_A(K_1, N)$, $\varphi \mapsto \varphi|_{K_1}$. Hence $\operatorname{Ext}_A^1(M, N)$ is the cokernel.

ii. Prove that $\operatorname{Ext}_A^h(M, N) \cong \operatorname{Ext}_A^{h-1}(K_1, N)$ for $h \geq 2$.

Solution: The shifted complex $\cdots \to P_3 \to P_2 \to P_1 \xrightarrow{d_1} K_1 \to 0$ is a projective resolution of K_1, so $\operatorname{Ext}_A^{h-1}(K_1, N)$ is the cohomology of $\operatorname{Hom}_A(P_{\bullet+1}, N)$ in degree $h-1$. For $h \geq 2$, the cohomology of $\operatorname{Hom}_A(P_\bullet, N)$ in degree h uses $\operatorname{Hom}_A(P_{h-1}, N) \to \operatorname{Hom}_A(P_h, N) \to \operatorname{Hom}_A(P_{h+1}, N)$, which is the same piece of the complex, so $\operatorname{Ext}_A^h(M, N) \cong \operatorname{Ext}_A^{h-1}(K_1, N)$.

Exercise 2.

(1) Let $S = \bigoplus_{i=1}^{4} \Bbbk e_i$ and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c \oplus \Bbbk d$ be the S-bimodule with

$$\Bbbk a = e_1 V e_2, \qquad \Bbbk b = e_2 V e_4, \qquad \Bbbk c = e_1 V e_3, \qquad \Bbbk d = e_3 V e_4,$$

and $e_i V e_j = 0$ for all other (i, j). For $\lambda \in \Bbbk$ put $A_\lambda = T_S(V)/(ab - \lambda cd)$.

i. Check that

$$h_{A_\lambda}(t) = \begin{pmatrix} 1 & t & t & t^2 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solution: The projectives are $e_1 A = \operatorname{span}\{e_1, a, c, cd\}$ (with $ab = \lambda cd$), $e_2 A = \{e_2, b\}$, $e_3 A = \{e_3, d\}$, $e_4 A = \{e_4\}$, which gives the Hilbert series.

ii. Find $R^\perp$ and a presentation of $A_\lambda^!$, for every $\lambda \in \Bbbk$.

Solution: $D(V)$ has the dual basis $a^* \in e_2 D(V) e_1$, $b^* \in e_4 D(V) e_2$, $c^* \in e_3 D(V) e_1$, $d^* \in e_4 D(V) e_3$, and the composable words of length 2 are $b^* a^*$, $d^* c^*$. Evaluating by $(\lambda \otimes \mu)(v \otimes w) = \sum_i \mu(v e_i)\, \lambda(e_i w)$: $(b^* \otimes a^*)(a \otimes b) = 1$ and it vanishes on $c \otimes d$; $(d^* \otimes c^*)(c \otimes d) = 1$ and it vanishes on $a \otimes b$. So

$$(\mu b^* a^* + \nu d^* c^*)(ab - \lambda cd) = \mu - \lambda\nu,$$

giving $R^\perp = \Bbbk(\lambda b^* a^* + d^* c^*)$ and $A_\lambda^! = T_S(D(V))/(\lambda b^* a^* + d^* c^*)$.

iii. Compute $(R^\perp)^\perp$ and verify that $(A_\lambda^!)^! \cong A_\lambda$.

Solution: Identifying $D(D(V)) = V$ (with $a^{**} = a$, and so on), the same computation shows $(\mu\, ab + \nu\, cd)(\lambda b^* a^* + d^* c^*) = \lambda\mu + \nu$, so the annihilator of $R^\perp$ is $\Bbbk(ab - \lambda cd) = R$ and $(A_\lambda^!)^! = A_\lambda$.

iv. Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are either both zero or both non-zero.

Solution: A graded isomorphism $\varphi : A_\lambda \to A_\mu$ over S fixes each e_i, so φ_1 preserves each $e_i V e_j$ (all at most one-dimensional):

$$\varphi(a) = \alpha a, \quad \varphi(b) = \beta b, \quad \varphi(c) = \gamma c, \quad \varphi(d) = \delta d, \qquad \alpha, \beta, \gamma, \delta \in \Bbbk^\times.$$

The induced map on $T_S(V)$ sends $ab - \lambda cd$ to $\alpha\beta\, ab - \lambda\gamma\delta\, cd$, which must lie in $\Bbbk(ab - \mu cd)$:

$$\mu\, \alpha\beta = \lambda\, \gamma\delta.$$

Hence $\lambda = 0 \Leftrightarrow \mu = 0$. Conversely if $\lambda, \mu \neq 0$ take $\alpha = \beta = \gamma = 1$, $\delta = \mu/\lambda$ (the inverse is of the same form); for $\lambda = \mu = 0$ take the identity.

v. Find the minimal graded projective resolution of each simple module $e_i S$.

Solution: $e_4S = e_4A$ is projective. For e_2S: the kernel of $e_2A \to e_2S$ is $\Bbbk b$, and $b\,e_4 = b$, $bA_+ = 0$, so it is isomorphic to $e_4A(-1)$ and $0 \to e_4A(-1) \to e_2A \to e_2S \to 0$ is the minimal resolution; similarly $0 \to e_4A(-1) \to e_3A \to e_3S \to 0$. For e_1S: the kernel of $e_1A \to e_1S$ is $e_1A_+ = \operatorname{span}\{a, c, cd\}$, and

$$d_1 : e_2A(-1) \oplus e_3A(-1) \to e_1A_+, \qquad (x, y) \mapsto ax + cy,$$

is surjective ($e_2 \mapsto a$, $e_3 \mapsto c$, $d \mapsto cd$, $b \mapsto \lambda cd$). Its kernel is one-dimensional, spanned by $z = (b, -\lambda d)$ in degree 2, and $zA_+ = 0$, so it is isomorphic to $e_4A(-2)$ (projective). So

$$0 \to e_4A(-2) \to e_2A(-1) \oplus e_3A(-1) \to e_1A \to e_1S \to 0$$

is the minimal resolution (the kernels lie in PA_+).

vi. Deduce that A_λ is Koszul for every $\lambda \in \Bbbk$.

Solution: In every resolution P_h is generated in degree h, so all are linear, hence so is the resolution of $S = \bigoplus_i e_iS$, and A_λ is Koszul.

vii. Compute $h_{A^!_\lambda}(t)$.

Solution: The only composable words of length 2 in $D(V)$ are b^*a^* and d^*c^*, both in $e_4A^!e_1$, and the relation $\lambda b^*a^* + d^*c^* = 0$ leaves $\dim e_4A^!_2e_1 = 1$ for every λ; there are no composable words of length 3. Hence $h_{A^!_\lambda}(t) = I + tN^{\mathsf{T}} + t^2E_{41}$, where $N = E_{12} + E_{13} + E_{24} + E_{34}$ and $h_{A_\lambda}(t) = I + tN + t^2E_{14}$.

viii. Show that $h_{A_\lambda}(t)\, h_{A^!_\lambda}(-t)^{\mathsf{T}} = I_4$.

Solution: Using $N^2 = 2E_{14}$ and $NE_{14} = E_{14}N = E_{14}^2 = 0$,

$$(I + tN + t^2E_{14})(I - tN + t^2E_{14}) = I + t^2(E_{14} - N^2 + E_{14}) = I.$$

(2) Let M be a graded right module with $M = M_mA$ for some $m \in \mathbb{Z}$, bounded below and locally finite-dimensional, with minimal graded projective resolution $P_\bullet \to M$.

i. Prove that $(P_h)_i = 0$ for $i < m + h$.

Solution: For $h = 0$: $M_i = 0$ for $i < m$ and $M = M_mA$, so $\operatorname{grtop} M$ is concentrated in degree m, and $\operatorname{grtop} P_0 \cong \operatorname{grtop} M$, so P_0 is generated in degree m and $(P_0)_i = 0$ for $i < m$. Let $h \geq 1$ and assume $(P_{h-1})_i = 0$ for $i < m + h - 1$. Then $P_{h-1}A_+$ vanishes in degrees $< m + h$, and so does $\operatorname{Im} d_h \subseteq P_{h-1}A_+$. Since $P_h \to \operatorname{Im} d_h$ is a projective cover, $\operatorname{grtop} P_h \cong \operatorname{grtop} \operatorname{Im} d_h$ is concentrated in degrees $\geq m + h$, so P_h is generated in degrees $\geq m + h$ and $(P_h)_i = 0$ for $i < m + h$.

ii. Deduce that $\operatorname{ext}^h(M, L(j)) = 0$ for $j > -m - h$ and every simple graded S-module L in degree 0.

Solution: Finally $\operatorname{ext}^h(M, L(j)) \cong \operatorname{Hom}_S((\operatorname{grtop} P_h)_{-j}, L) = 0$ whenever $-j < m + h$, that is, $j > -m - h$.

Exercise 3.

(1) Let $n \geq 3$ and $Z \subseteq \{1, \dots, n-2\}$, and let $A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$ be the quotient of the algebra of upper triangular $n \times n$ matrices, with $\deg e_{ij} = j - i$.

i. Check that $h_{A_Z}(t)_{ij} = t^{j-i}$ if $i \leq j$ and there is no $k \in Z$ with $i \leq k \leq j-2$, and 0 otherwise.

Solution: $e_{ij} = e_{ik}e_{k,k+2}e_{k+2,j}$ for $i \leq k$, $k+2 \leq j$, so the ideal is spanned by the e_{ij} for which such $k \in Z$ exists (it is a two-sided ideal, since a product $e_{pq}e_{ij}e_{rs} \neq 0$ has $p \leq i$ and $j \leq s$). The remaining matrix units form a basis of A_Z, and $e_{ij} \in e_iA_Ze_j$ has degree $j - i$.

ii. For each i, find the minimal graded projective resolution of $e_{ii}S$.
Hint. Let $\ell_i \geq 0$ be maximal with $i, i+1, \dots, i+\ell_i - 1 \in Z$. Show that $P_h = e_{i+h,i+h}A(-h)$.

Solution: $e_{nn}S = e_{nn}A$ is projective. Let $i < n$. The kernel of $e_{ii}A \to e_{ii}S$ is $e_{ii}A_+ = e_{i,i+1}A$, since $e_{ij} = e_{i,i+1}e_{i+1,j}$. The map $e_{i+1,i+1}A(-1) \to e_{i,i+1}A$, $x \mapsto e_{i,i+1}x$, is surjective, and its kernel is spanned by the non-zero $e_{i+1,j}$ with $e_{ij} = 0$. By (a), $e_{ij} = 0 \neq e_{i+1,j}$ happens exactly when $i \in Z$ and $j \geq i+2$. So the kernel is 0 if $i \notin Z$, and equals $e_{i+1,i+1}A_+$ if $i \in Z$. In the second case we are in the same situation at $i+1$. By induction, the minimal resolution is

$$0 \to e_{i+\ell_i+1,i+\ell_i+1}A(-\ell_i - 1) \to \cdots \to e_{i+1,i+1}A(-1) \to e_{ii}A \to e_{ii}S \to 0,$$

that is, $P_h = e_{i+h,i+h}A(-h)$ for $0 \leq h \leq \ell_i + 1$, and $P_h = 0$ beyond (all kernels lie in PA_+).

iii. Deduce that A_Z is Koszul.

Solution: It is linear, so A_Z is Koszul.

iv. Compute $\dim_{\Bbbk} \operatorname{ext}^h(e_{ii}S, e_{jj}S(m))$ for all h, j, m, with ℓ_i as in the hint of (b).

Solution: By the description of ext with simple coefficients via grtop P_h, $\dim \operatorname{ext}^h(e_{ii}S, e_{jj}S(m)) = 1$ if $j = i + h$, $m = -h$ and $h \leq \ell_i + 1$ (and $i < n$ if $h > 0$), and 0 otherwise.

v. Let $n = 4$ and $Z = \{1\}$. Then $A_Z = T_S(V)/(R)$ with $S = \bigoplus_i \Bbbk e_{ii}$, $V = \Bbbk e_{12} \oplus \Bbbk e_{23} \oplus \Bbbk e_{34}$ and $R = \Bbbk\,(e_{12} \otimes e_{23})$. Compute $R^\perp$ and a presentation of $A_Z^!$.

Solution: Write $a = e_{12}$, $b = e_{23}$, $c = e_{34}$; a basis of A_Z is e_{ii}, a, b, c, bc, so $h_{A_Z}(t) = I + t(E_{12}+E_{23}+E_{34}) + t^2E_{24}$. $D(V)$ has the dual basis $a^* \in e_{22}D(V)e_{11}$, $b^* \in e_{33}D(V)e_{22}$, $c^* \in e_{44}D(V)e_{33}$; the composable words of length 2 are b^*a^* and c^*b^*. With $(\lambda \otimes \mu)(v \otimes w) = \sum_i \mu(ve_i)\,\lambda(e_iw)$,

$$(b^* \otimes a^*)(a \otimes b) = 1, \qquad (c^* \otimes b^*)(a \otimes b) = 0,$$

so $R^\perp = \Bbbk\, c^*b^*$ and $A_Z^! = T_S(D(V))/(c^*b^*)$.

vi. Find a basis of $A_Z^!$ and compute $h_{A_Z^!}(t)$.

Solution: A word of length 3 would be $c^*b^*a^*$, which contains c^*b^*; hence a basis of $A^!_Z$ is $e_{ii},\ a^*,\ b^*,\ c^*,\ b^*a^*$, and

$$h_{A^!_Z}(t) = I + t(E_{21} + E_{32} + E_{43}) + t^2E_{31}.$$

vii. Verify that $h_{A_Z}(t)\, h_{A^!_Z}(-t)^{\mathsf{T}} = I_4$.

Solution: With $N = E_{12} + E_{23} + E_{34}$ we have $N^2 = E_{13} + E_{24}$ and $NE_{13} = E_{24}N = E_{24}E_{13} = 0$, so

$$h_{A_Z}(t)h_{A^!_Z}(-t)^{\mathsf{T}} = (I + tN + t^2E_{24})(I - tN + t^2E_{13}) = I + t^2\big(E_{13} - N^2 + E_{24}\big) = I.$$

(2) Let $A = \Bbbk\langle x, y\rangle/(yx,\ x^2 + y^2)$ with $\deg x = \deg y = 1$.

i. Show that $x^3 = 0$ and $y^3 = 0$, and deduce that $A_d = 0$ for $d \geq 3$ and $h_A(t) = 1 + 2t + 2t^2$.
Hint. $x^3 = x \cdot x^2 = -xy^2$ and $x^3 = x^2 \cdot x = -y^2x = -y(yx)$, and similarly for the other words of length 3.

Solution: We have $x^3 = x \cdot x^2 = -xy^2$ and $x^3 = x^2 \cdot x = -y^2x = -y(yx) = 0$, so $x^3 = 0$ and $xy^2 = 0$. Likewise $y^3 = y \cdot y^2 = -yx^2 = -(yx)x = 0$. The remaining words of length 3 contain yx or are $xxy = -y^3 = 0$ and $xyy = -x^3 = 0$, hence vanish. So $A_3 = 0$, and $A_d = 0$ for $d \geq 3$ since A is generated in degree 1. Also $\dim A_1 = 2$ and $\dim A_2 = 4 - 2 = 2$, so $h_A(t) = 1 + 2t + 2t^2$.

ii. Compute the first three terms of the minimal graded projective resolution $P_\bullet \to \Bbbk$

iii. Show that A is not Koszul, but quadratic.

Solution: We have

$$cA(-2)b \oplus dA(-2) \to aA(-1) \oplus bA(-1) \to A \to \Bbbk \to 0$$

with basis vector a, b, c, d so that

$$d_1 : \begin{array}{l} a \mapsto x \\ b \mapsto y \end{array}, \quad d_2 : \begin{array}{c} c \mapsto ax - by \\ d \mapsto bx \end{array}.$$

Then $\operatorname{Ker}(d_2)$ is spanned by cxy, cx^2, dxy, dx^2.

(3) Let A be a graded algebra with $A_0 = \Bbbk$ and let M be a bounded below locally finite-dimensional graded A-module whose minimal graded projective resolution $P_\bullet \to M$ has $P_h = 0$ for $h \gg 0$. Prove that

$$h_M(t) = h_A(t) \cdot \sum_{h \geq 0} (-1)^h \sum_{i \in \mathbb{Z}} \dim_\Bbbk \operatorname{ext}^h_A(M, \Bbbk(-i))\, t^i.$$

Solution: By the description of ext via a minimal resolution, with $S = \Bbbk$, $\dim \operatorname{ext}^h_A(M, \Bbbk(-i)) = \dim(\operatorname{grtop} P_h)_i$, so the inner sum is $h_{\operatorname{grtop} P_h}(t)$. Since $P_h \cong \operatorname{grtop} P_h \otimes_\Bbbk A$ as graded vector

spaces, $h_{P_h}(t) = h_{\mathrm{grtop}\, P_h}(t)\, h_A(t)$. Taking the degree i component of the exact sequence $0 \to P_N \to \cdots \to P_0 \to M \to 0$ of vector spaces and the alternating sum of dimensions gives $h_M(t) = \sum_h (-1)^h h_{P_h}(t)$, which is the claim.

Exercise 4.

(1) Let $n \geq 2$ and $A = \Bbbk[x]/(x^n)$ with $\deg x = 1$.

i. Find the minimal graded projective resolution of $\Bbbk = A/A_+$.

Solution: Let g_h denote the generator of P_h. $P_0 = A$ and $\operatorname{Ker} d_0 = xA$, so $P_1 = A(-1)$ with $d_1(g_1) = x$. Since $xa = 0$ if and only if $a \in x^{n-1}A$, $\operatorname{Ker} d_1 = x^{n-1}g_1A$ is generated in degree n, so $P_2 = A(-n)$ with $d_2(g_2) = x^{n-1}g_1$. Since $x^{n-1}a = 0$ if and only if $a \in xA$, $\operatorname{Ker} d_2 = xg_2A$ is generated in degree $n+1$, so $P_3 = A(-n-1)$ with $d_3(g_3) = xg_2$, and the pattern repeats with period 2:

$$P_{2m} = A(-nm), \qquad P_{2m+1} = A(-nm-1) \qquad (m \geq 0).$$

All differentials map into PA_+ since $n \geq 2$, so the resolution is minimal.

ii. Compute $\operatorname{ext}^h_A(\Bbbk, \Bbbk(j))$ for all h, j.

Solution: Here $\operatorname{grtop} P_h = \Bbbk$ is concentrated in degree nm for $h = 2m$ and in degree $nm+1$ for $h = 2m+1$, so $\operatorname{ext}^h_A(\Bbbk, \Bbbk(j)) \cong \operatorname{Hom}_S((\operatorname{grtop} P_h)_{-j}, \Bbbk)$ is $\Bbbk$ if $(h, j) = (2m, -nm)$ or $(2m+1, -nm-1)$ for some $m \geq 0$, and 0 otherwise.

iii. Let $1 \leq r < n$ and $M = A/x^rA$. Find the minimal graded projective resolution of M.

Solution: $\operatorname{Ker}(A \to M) = x^rA$ is generated by x^r in degree r, so $P_1 = A(-r)$ with $d_1(g_1) = x^r$. Since $x^ra = 0$ if and only if $a \in x^{n-r}A$, $\operatorname{Ker} d_1 = x^{n-r}g_1A$ is generated in degree n, so $P_2 = A(-n)$ with $d_2(g_2) = x^{n-r}g_1$. Since $x^{n-r}a = 0$ if and only if $a \in x^rA$, $\operatorname{Ker} d_2 = x^rg_2A$ is generated in degree $n+r$, so $P_3 = A(-n-r)$, and the pattern repeats with period 2:

$$P_{2m} = A(-nm), \qquad P_{2m+1} = A(-nm-r) \qquad (m \geq 0).$$

All differentials map into PA_+ since $1 \leq r < n$, so the resolution is minimal.

iv. Compute $\operatorname{ext}^h_A(M, \Bbbk(j))$ for all h, j.

Solution: As in (b), $\operatorname{ext}^h_A(M, \Bbbk(j))$ is $\Bbbk$ if $(h, j) = (2m, -nm)$ or $(2m+1, -nm-r)$ for some $m \geq 0$, and 0 otherwise.

(2) Let M be a bounded below locally finite-dimensional graded module with minimal graded projective resolution $P_\bullet \to M$, and let $n \geq 0$. Prove that the following are equivalent:

(a) $P_h = 0$ for all $h > n$;

(b) $\operatorname{ext}^{n+1}(M, L(j)) = 0$ for every simple graded S-module L in degree 0 and every $j \in \mathbb{Z}$.

Hint. Use the description of $\operatorname{ext}^h(M, L(j))$ via $\operatorname{grtop} P_h$ and the graded Nakayama lemma.

Solution: (a)⇒(b): $\operatorname{ext}^{n+1}(M, L(j)) \cong \operatorname{Hom}_S((\operatorname{grtop} P_{n+1})_{-j}, L) = 0$, because $P_{n+1} = 0$.
(b)⇒(a): By the same description, the hypothesis says $\operatorname{Hom}_S((\operatorname{grtop} P_{n+1})_{-j}, L) = 0$ for all simple L and all j. Every non-zero S-module has a non-zero map to some simple module, since S is semisimple, so $\operatorname{grtop} P_{n+1} = 0$, that is, $P_{n+1} = P_{n+1}A_+$. As P_{n+1} is bounded below, graded Nakayama gives $P_{n+1} = 0$. Then $\operatorname{Ker} d_{n+1} = 0$, so $\operatorname{Im} d_{n+2} = 0$, and $P_{n+2} \to$

$\operatorname{Im} d_{n+2} = 0$ has kernel $P_{n+2} \subseteq P_{n+2}A_+$ by minimality, so $P_{n+2} = 0$ again by Nakayama. Induction gives $P_h = 0$ for all $h > n$.

(3) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = \Bbbk a \oplus \Bbbk b$ with $a \in e_1 V e_2$, $b \in e_2 V e_1$, and $A = T_S(V)/(a \otimes b)$.

i. Check that $h_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}$.

Solution: $ab = 0$, so $e_1A = \operatorname{span}\{e_1, a\}$ and $e_2A = \operatorname{span}\{e_2, b, ba\}$, with $a \in e_1Ae_2$, $b \in e_2Ae_1$, $ba \in e_2Ae_2$.

ii. Find the minimal graded projective resolutions of e_1S and e_2S.

Solution: The kernel of $e_2A \to e_2S$ is $\operatorname{span}\{b, ba\} = bA$, and $e_1A(-1) \to bA$, $x \mapsto bx$, is a degree-preserving bijection ($e_1 \mapsto b$, $a \mapsto ba$). So $0 \to e_1A(-1) \to e_2A \to e_2S \to 0$ is the minimal resolution of e_2S. The kernel of $e_1A \to e_1S$ is $\Bbbk a$, and $aA_+ = 0$ (as $ab = 0$), so it is isomorphic to $e_2S(-1)$; splicing the resolution of e_2S shifted by 1 gives

$$0 \to e_1A(-2) \to e_2A(-1) \to e_1A \to e_1S \to 0.$$

iii. Deduce that A is Koszul.

Solution: Both resolutions are linear, hence so is that of $S = e_1S \oplus e_2S$, and A is Koszul.

iv. Compute a presentation of $A^!$.

Solution: The composable words of length 2 in V are $a \otimes b$ and $b \otimes a$. Let $a^* \in e_2D(V)e_1$, $b^* \in e_1D(V)e_2$ be the dual basis. The functional $\theta(a^* \otimes b^*)$ is dual to $b \otimes a$ and vanishes on $a \otimes b$, so $R^\perp = \Bbbk\, a^*b^*$ and $A^! = T_S(D(V))/(a^*b^*)$.

v. Find a basis of $A^!$ and compute $h_{A^!}(t)$.

Solution: Since $a^*b^* = 0$, a basis is $e_1, e_2, a^*, b^*, b^*a^*$, so

$$h_{A^!}(t) = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix}.$$

vi. Verify that $h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I_2$.

Solution:

$$h_A(t) h_{A^!}(-t)^{\mathsf{T}} = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix} \begin{pmatrix} 1+t^2 & -t \\ -t & 1 \end{pmatrix} = I_2.$$